HZM-N Series

Silicon Epitaxial Planar Zener Diode for Stabilizer

R07DS0358EJ0600
Rev.6.00
May 19, 2011

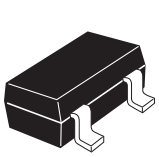
Features

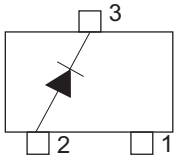
- Wide spectrum from 1.9 V through 38 V of zener voltage provide flexible application.
- MPAK Package is suitable for high density surface mounting and high speed assembly.

Ordering Information

Part No	Laser Mark	Package Name	Package Code	Taping Abbreviation (Quantity)
HZM-N Series TL HZM-N Series TR	Let to Mark Code	MPAK	PLSP0003ZC-A	TL (3,000pcs / reel) TR (3,000pcs / reel)

Pin Arrangement





(Top View)

1. NC
2. Anode
3. Cathode

Absolute Maximum Ratings

(Ta = 25°C)

Item	Symbol	Value	Unit
Power dissipation	Pd *[1]	200	mW
Junction temperature	Tj	150	°C
Storage temperature	Tstg	–55 to +150	°C

Note: 1. See Fig. 3.

Electrical Characteristics

(Ta = 25°C)

Type	Grade	Zener Voltage			Reverse Current		Dynamic Resistance	
		V_Z (V)*[1]		Test Condition	I_R (μA)	Test Condition	r_d (Ω)	Test Condition
		Min	Max	I_Z (mA)	Max	V_R (V)	Max	I_Z (mA)
HZM2.0N	B	1.90	2.20	5	120	0.5	100	5
HZM2.2N	B	2.10	2.40	5	120	0.7	100	5
HZM2.4N	B	2.30	2.60	5	120	1.0	100	5
HZM2.7N	B1	2.50	2.75	5	120	1.0	110	5
	B2	2.65	2.90					
HZM3.0N	B1	2.80	3.05	5	50	1.0	120	5
	B2	2.95	3.20					
HZM3.3N	B1	3.10	3.35	5	20	1.0	130	5
	B2	3.25	3.50					
HZM3.6N	B1	3.40	3.65	5	10	1.0	130	5
	B2	3.55	3.80					
HZM3.9N	B1	3.70	3.97	5	10	1.0	130	5
	B2	3.87	4.10					
HZM4.3N	B1	4.01	4.21	5	10	1.0	130	5
	B2	4.15	4.34					
	B3	4.28	4.48					
HZM4.7N	B1	4.42	4.61	5	10	1.0	130	5
	B2	4.55	4.75					
	B3	4.69	4.90					
HZM5.1N	B1	4.84	5.04	5	5	1.5	130	5
	B2	4.98	5.20					
	B3	5.14	5.37					
HZM5.6N	B1	5.31	5.55	5	5	2.5	80	5
	B2	5.49	5.73					
	B3	5.67	5.92					
HZM6.2N	B1	5.86	6.12	5	2	3.0	50	5
	B2	6.06	6.33					
	B3	6.26	6.53					
HZM6.8N	B1	6.47	6.73	5	2	3.5	30	5
	B2	6.65	6.93					
	B3	6.86	7.14					
HZM7.5N	B1	7.06	7.36	5	2	4.0	30	5
	B2	7.28	7.60					
	B3	7.52	7.84					

Note: 1. Tested with pulse (Pw = 40 ms)

Type	Grade	Zener Voltage			Reverse Current		Dynamic Resistance	
		V_Z (V)*[1]		Test Condition	I_R (μA)	Test Condition	r_d (Ω)	Test Condition
		Min	Max	I_Z (mA)	Max	V_R (V)	Max	I_Z (mA)
HZM8.2N	B1	7.76	8.10	5	2	5.0	30	5
	B2	8.02	8.36					
	B3	8.28	8.64					
HZM9.1N	B1	8.56	8.93	5	2	6.0	30	5
	B2	8.85	9.23					
	B3	9.15	9.55					
HZM10N	B1	9.45	9.87	5	2	7.0	30	5
	B2	9.77	10.21					
	B3	10.11	10.55					
HZM11N	B1	10.44	10.88	5	2	8.0	30	5
	B2	10.76	11.22					
	B3	11.10	11.56					
HZM12N	B1	11.42	11.90	5	2	9.0	35	5
	B2	11.74	12.24					
	B3	12.08	12.60					
HZM13N	B1	12.47	13.03	5	2	10.0	35	5
	B2	12.91	13.49					
	B3	13.37	13.96					
HZM15N	B1	13.84	14.46	5	2	11.0	40	5
	B2	14.34	14.98					
	B3	14.85	15.52					
HZM16N	B1	15.37	16.01	5	2	12.0	40	5
	B2	15.85	16.51					
	B3	16.35	17.09					
HZM18N	B1	16.94	17.70	5	2	13.0	45	5
	B2	17.56	18.35					
	B3	18.21	19.03					
HZM20N	B1	18.86	19.70	5	2	15.0	50	5
	B2	19.52	20.39					
	B3	20.21	21.08					
HZM22N	B1	20.88	21.77	5	2	17.0	55	5
	B2	21.54	22.47					
	B3	22.23	23.17					
HZM24N	B1	22.93	23.96	5	2	19.0	60	5
	B2	23.72	24.78					
	B3	24.54	25.57					
HZM27N	B	25.10	28.90	2	2	21.0	70	2
HZM30N	B	28.00	32.00	2	2	23.0	80	2
HZM33N	B	31.00	35.00	2	2	25.0	80	2
HZM36N	B	34.00	38.00	2	2	27.0	90	2

Note: 1. Tested with pulse (P_W = 40 ms)

Mark Code

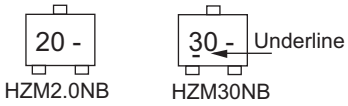
Type	Grade	Mark No.
HZM2.0N	B	2 0 –
HZM2.2N	B	2 2 –
HZM2.4N	B	2 4 –
HZM2.7N	B1	2 7 1
	B2	2 7 2
HZM3.0N	B1	3 0 1
	B2	3 0 2
HZM3.3N	B1	3 3 1
	B2	3 3 2
HZM3.6N	B1	3 6 1
	B2	3 6 2
HZM3.9N	B1	3 9 1
	B2	3 9 2
HZM4.3N	B1	4 3 1
	B2	4 3 2
	B3	4 3 3
HZM4.7N	B1	4 7 1
	B2	4 7 2
	B3	4 7 3
HZM5.1N	B1	5 1 1
	B2	5 1 2
	B3	5 1 3
HZM5.6N	B1	5 6 1
	B2	5 6 2
	B3	5 6 3

Type	Grade	Mark No.
HZM6.2N	B1	6 2 1
	B2	6 2 2
	B3	6 2 3
HZM6.8N	B1	6 8 1
	B2	6 8 2
	B3	6 8 3
HZM7.5N	B1	7 5 1
	B2	7 5 2
	B3	7 5 3
HZM8.2N	B1	8 2 1
	B2	8 2 2
	B3	8 2 3
HZM9.1N	B1	9 1 1
	B2	9 1 2
	B3	9 1 3
HZM10N	B1	1 0 1
	B2	1 0 2
	B3	1 0 3
HZM11N	B1	1 1 1
	B2	1 1 2
	B3	1 1 3
HZM12N	B1	1 2 1
	B2	1 2 2
	B3	1 2 3

Type	Grade	Mark No.
HZM13N	B1	1 3 1
	B2	1 3 2
	B3	1 3 3
HZM15N	B1	1 5 1
	B2	1 5 2
	B3	1 5 3
HZM16N	B1	1 6 1
	B2	1 6 2
	B3	1 6 3
HZM18N	B1	1 8 1
	B2	1 8 2
	B3	1 8 3
HZM20N	B1	2 0 1
	B2	2 0 2
	B3	2 0 3
HZM22N	B1	2 2 1
	B2	2 2 2
	B3	2 2 3
HZM24N	B1	2 4 1
	B2	2 4 2
	B3	2 4 3
HZM27N	B	2 7 –
HZM30N	B	3 0 –
HZM33N	B	3 3 –
HZM36N	B	3 6 –

Example of Marking

1. One grade type (grade type B)



HZM2.0NB HZM30NB

2. Two grade type (B1, B2)



HZM3.0NB1 HZM3.0NB2

3. Three grade type (B1, B2, B3)



HZM4.3NB1 HZM4.3NB2 HZM4.3NB3

Notes: 1. Ordering P/N HZM-N series are delivered taped (TL/TR).
2. Choose one taping code and adhere to parts No.
Example: HZM2.0NBTL (or TR), HZM2.2NBTL (or TR), HZM36NBTL (or TR).
(Grade B type)
HZM2.7NB1TL (or TR), HZM2.7NB2TL (or TR), HZM24NB3TL (or TR).
(Grade B1, B2, B3 type)

Main Characteristics

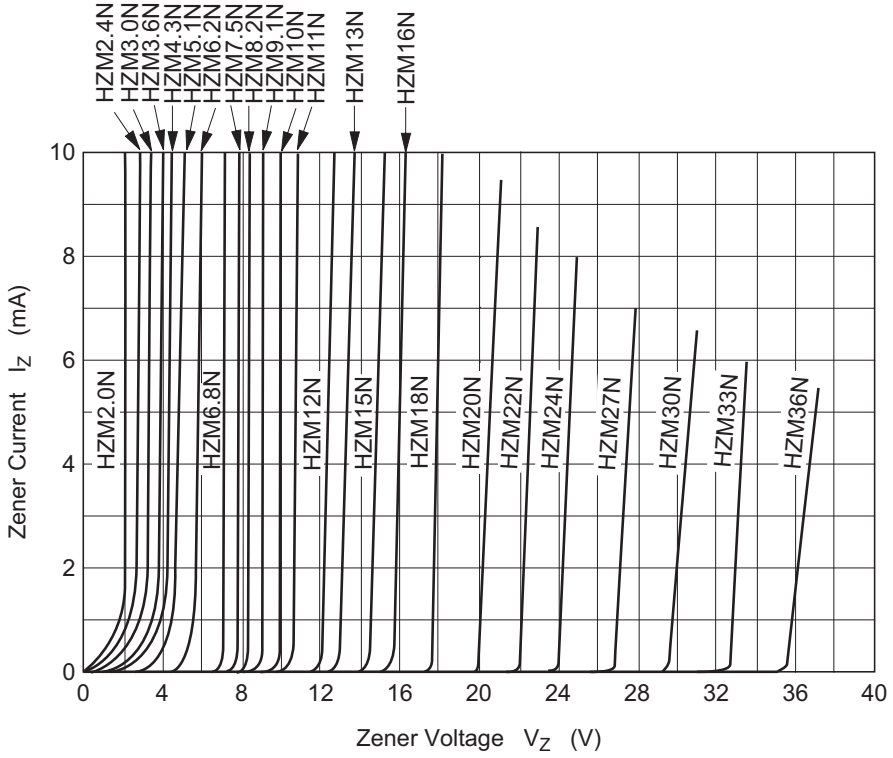


Fig.1 Zener current vs. Zener voltage

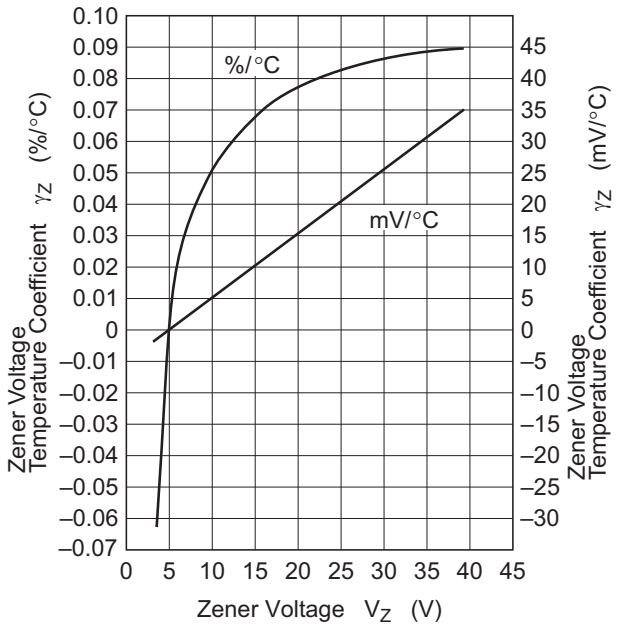


Fig.2 Temperature Coefficient vs. Zener voltage

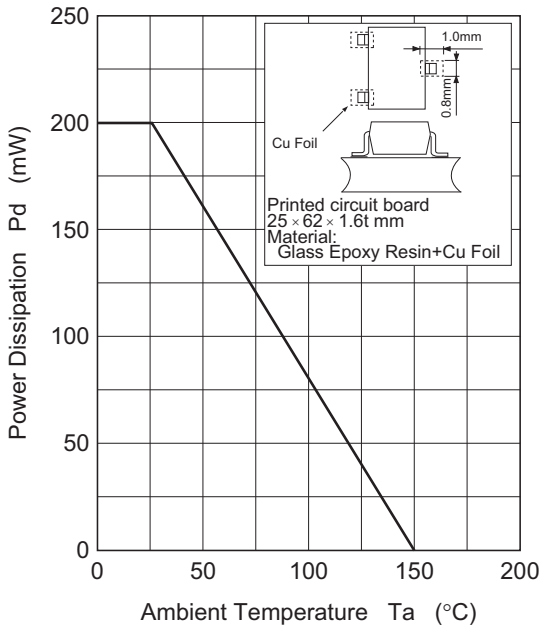
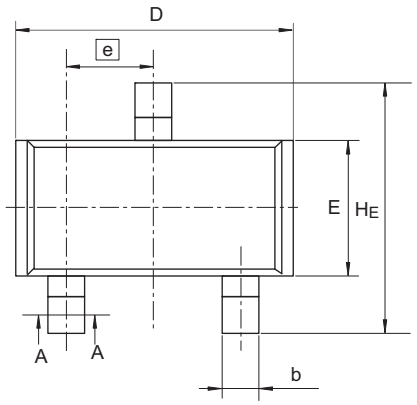
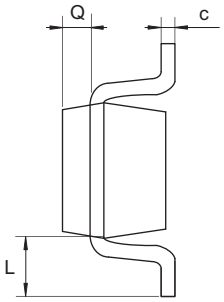


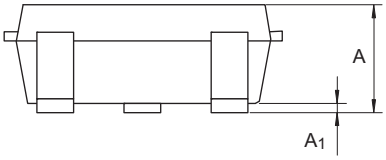
Fig.3 Power Dissipation vs. Ambient Temperature

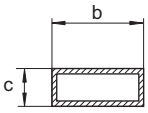
Package Dimensions

Package Name	JEITA Package Code	RENESAS Code	Previous Code	MASS[Typ.]
MPAK	SC-59A	PLSP0003ZC-A	MPAK(D) / MPAK(D)V	0.011g

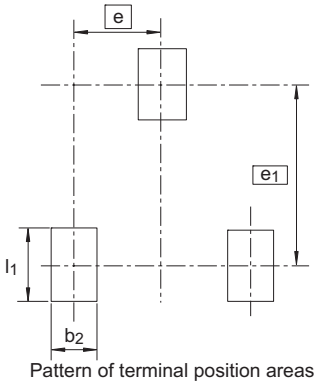








A — A Section



Pattern of terminal position areas

Reference Symbol	Dimension in Millimeters		
	Min	Nom	Max
A	1.0	-	1.3
A_1	0	-	0.1
b	0.35	0.4	0.5
c	0.1	0.16	0.26
D	2.7	-	3.1
E	1.35	1.5	1.65
e	-	0.95	-
H_E	2.2	2.8	3.0
L	-	0.65	-
b_2	-	-	0.55
e_1	-	1.95	-
l_1	-	-	1.05
Q	-	0.3	-

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